

15/09/23

Maths

$$\theta 1 - (3) \quad 3$$

$$\theta 2 - \tan 60^\circ$$

$$\theta 3 - 100^\circ$$

$$\theta 4 - 10 \text{ cm}$$

$$\theta 5 - a = 0$$

$\theta 6 -$ Four decimal places

$$\theta 7 - 3 \text{ and } 1$$

$$\theta 8 - (0, -1)$$

$\theta 9 -$ No solution

$$\theta 10 - k = \frac{9}{2}$$

Q11- $K \neq 3$

Q12- 8

Q13- Similar

Q14- $u^3 y^4 z^4$

Q15- $u^2 - u + 12$

Q16- Intersecting & Consist Coincident

Q17- Parallel

Q18- 16:81

Q19- Both A & R are correct but R is not the correct explanation of A.

Q20 A is false but R is true.

Section - B

Q21- $6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$

$$= \begin{array}{r|l} 5 & 725 \\ \hline 5 & 145 \\ \hline 29 & 29 \\ \hline & 1 \end{array}$$

$$= 5 \times 5 \times 29$$

Since it is the product of the prime it is a composite number.

Q22- $4u + 6y = 7$, $ku + 18y = 21$

$$\frac{4}{k} = \frac{6}{18} = \frac{7}{21}$$

$$6k = 72$$

$$k = 12$$

k should be equal to 12

Q23 $(0,0)$, $(-6,8)$

$$\begin{aligned} \sqrt{u^2 + y^2} &= \sqrt{36 + 64} \\ &= 10 \end{aligned}$$

$$\text{Q24- } \sin 3A = \cos A - 10$$

$$= \cos 90 - 3A = \cos A - 10$$

$$\sin \theta = \cos 90^\circ - 0$$

$$90 - 3A = A - 10$$

$$90 + 10 = A + 3A$$

$$4A = 100$$

$$A = \frac{100}{4} = 25^\circ$$

$$\text{Q25 } \angle AOB = \angle DOC \text{ [Vertical Angles]}$$

$$\angle OAB = \angle ODC \text{ [Alternate Angles]}$$

$$\angle ABO = \angle DCO \text{ [" "]}$$

From A-A-A

$$\triangle AOB \sim \triangle DOC$$

Section-c

$$\text{Q26- } w.k.T$$

$$\cos(90 - \theta) = \sin \theta$$

$$\sin(90 - \theta) = \cos \theta$$

$$\tan(90 - \theta) = \cot \theta$$

$$= \sin(50^\circ + \theta) - \cos(40^\circ - \theta) + \tan 1^\circ \times \tan 10^\circ$$

$$\tan 20^\circ \times \tan 70^\circ \times \tan 80^\circ \times \tan 89^\circ$$

$$= \cos(40^\circ - \theta) - \cos(40^\circ - \theta) + \cot 89^\circ \times \cos 60^\circ \times$$

$$\cot 70^\circ \times \tan 70^\circ \times \tan 80^\circ \times \tan 89^\circ$$

$$= \cot 89^\circ \cdot \cot 80^\circ \cdot \cot 70^\circ \cdot \tan 70^\circ \cdot \tan 80^\circ \cdot \tan 89^\circ$$

$$= \cot \theta = \frac{1}{\tan \theta}$$

$$\cancel{\tan 70^\circ} \cdot \cancel{\tan 80^\circ} \cdot \cancel{\tan 89^\circ} = 1$$

$$\cancel{\tan 70^\circ} \cdot \cancel{\tan 80^\circ} \cdot \cancel{\tan 89^\circ}$$

Hence proved.

Q27- It states that if a line is parallel to the side of a triangle which intersects the other side into two distinct points then the line divides those sides of the triangle in proportion.

Let ABC be a triangle

To prove $\frac{AD}{DB} = \frac{AE}{EC}$

Join BE, CD

Draw $EF \perp AB$, $DG \perp CA$

Since $EF \perp AB$, $DG \perp CA$

EF is the height of triangle ADE & ABE

Area of triangle $\frac{1}{2} \times DB \times EF$

$$\frac{A_2(\triangle ADE)}{A_2(\triangle DBE)} = \frac{\frac{1}{2} \times AD \times EF}{\frac{1}{2} \times DB \times EF} = \frac{AD}{DB} \quad \text{--- (1)}$$

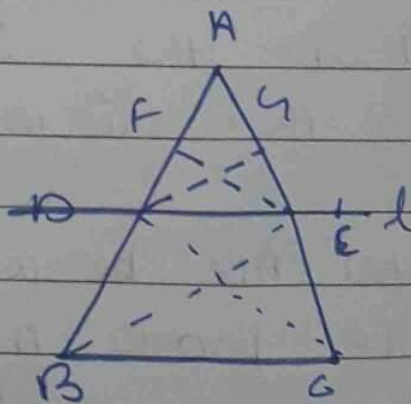
Similarly

$$\frac{A_2(\triangle ADE)}{A_2(\triangle DCE)} = \frac{\frac{1}{2} \times AE \times DG}{\frac{1}{2} \times EC \times DG} = \frac{AE}{EC} \quad \text{--- (2)}$$

$$A_2(\triangle DBE) = A_2(\triangle DCE) \quad \text{--- (3)}$$

From (1) (2) (3) we have

$$\frac{AD}{DB} = \frac{AE}{EC}$$



Q28- let Sagar Age be = x
 " Tiw " " = y

$$2(5-x) = (5-y)$$

$$10 - 2x = 5 - y$$

$$-2x + 5 + y$$

$$-2x + y + 5 = 0 \quad \text{--- (1)}$$

$$-2x + y = -5$$

$$(10+x)+10 = (10+y)$$

$$20+x = 10+y$$

$$x - y + 10 = 0 \quad \text{--- (2)}$$

$$x - y = -10$$

Comparing (1) & (2)

$$-2x + y + 5 = 0$$

$$x - y + 10 = 0$$

$$-x + 15 = 0$$

$$x = 15$$

$$x = 15$$

$$15 - y = -10$$

$$-y = -25$$

$$y = 25$$

Age of Sagar = 15

" " Tiw = 25

Q29- $A(-5, 7), B(4, 5), C(-1, -6), D(4, 5)$

Join A to C

$$\therefore \text{Ar}(\text{ABCD}) = \text{Ar}(\text{ABC}) + \text{Ar}(\text{ADC})$$

$$\text{Area of } \triangle ABC = \frac{1}{2} [-5(-5+6) - 4(-6-7) - 1(7+5)]$$

$$= \frac{1}{2} 35 \text{ sq. unit}$$

$$\begin{aligned} \text{Ar of } \triangle ADC &= \frac{1}{2} [-5(5+6) + (4(-6-7) - 1(7-5))] \\ &= \frac{1}{2} (-109) \quad \text{[Area cannot be Neg.]} \\ &= \frac{1}{2} 109 \text{ sq. unit} \end{aligned}$$

$$\text{Ar}(\text{ABCD}) = \frac{1}{2} 35 + \frac{1}{2} 109$$

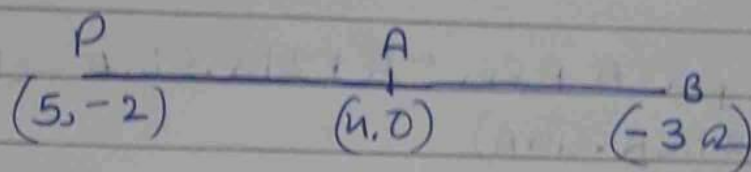
$$= \frac{1}{2} 144$$

$$= 72 \text{ sq. unit}$$

$$= 72 \text{ sq. unit}$$

Q30 $1353 = 861 \times 1 + 492$
 $861 = 492 \times 1 + 369$

Q31 Let the Point $P(u, 0)$ is equidistant from the pts.



i.e. $AP = BP$

$AP^2 = BP^2$

$(5-u)^2 + (-2-0)^2 = (-3-u)^2 + (2-0)^2$

$\Rightarrow u^2 + 25 - 10u + 4 = u^2 + 9 + 6u + 4$

$10u + 6u + 9 + 4 - u^2 - 25 = 0$

$16 - 16 = 0$

$u^2 - 16 = 0$
 16

So $(1, 0)$ is the point

$$\text{Q 32- } \tan A + \sin A = m$$

$$\tan A - \sin A = n$$

$$m^2 = (\tan A + \sin A)^2 = \tan^2 A + \sin^2 A + 2 \tan A \times \sin A$$

$$n^2 = (\tan A - \sin A)^2 = \tan^2 A + \sin^2 A - 2 \times \tan A \times \sin A$$

$$\therefore m^2 - n^2 = \tan^2 A + \sin^2 A + 2 \tan A \cdot \sin A - (\tan^2 A + \sin^2 A - 2 \tan A \cdot \sin A)$$

$$= 4 \tan A \cdot \sin A$$

$$mn = (\tan A + \sin A)(\tan A - \sin A)$$

$$= \tan^2 A - \sin^2 A = \frac{\sin^2 A}{\sec^2 A} = \sin^2 A$$

$$= \sin^2 A (\sec^2 A - 1)$$

$$\text{w.k.T}$$

$$\sec^2 A - \tan^2 A = 1$$

$$= \sin^2 A \times \tan^2 A = \sqrt{mn}$$

$$\text{LHS} = 4 \text{ RHS}$$

$$m^2 - n^2 = 4 \sqrt{mn}$$

033-

$$u - y + 1 = 0$$

$$y = u + 1$$

$$\text{Put } u = -1$$

$$y = 0$$

$$\text{Put } u = 1$$

$$y = 2$$

$$\text{Put } u = 2$$

$$y = 3$$

u	-1	1	2
y	0	2	3

$$3u + 2y - 12 = 0$$

$$2y = 12 - 3u + 12$$

$$y = \frac{-3u + 12}{2}$$

$$\text{Put } u = 0$$

$$y = 0 + \frac{12}{2} = 6$$

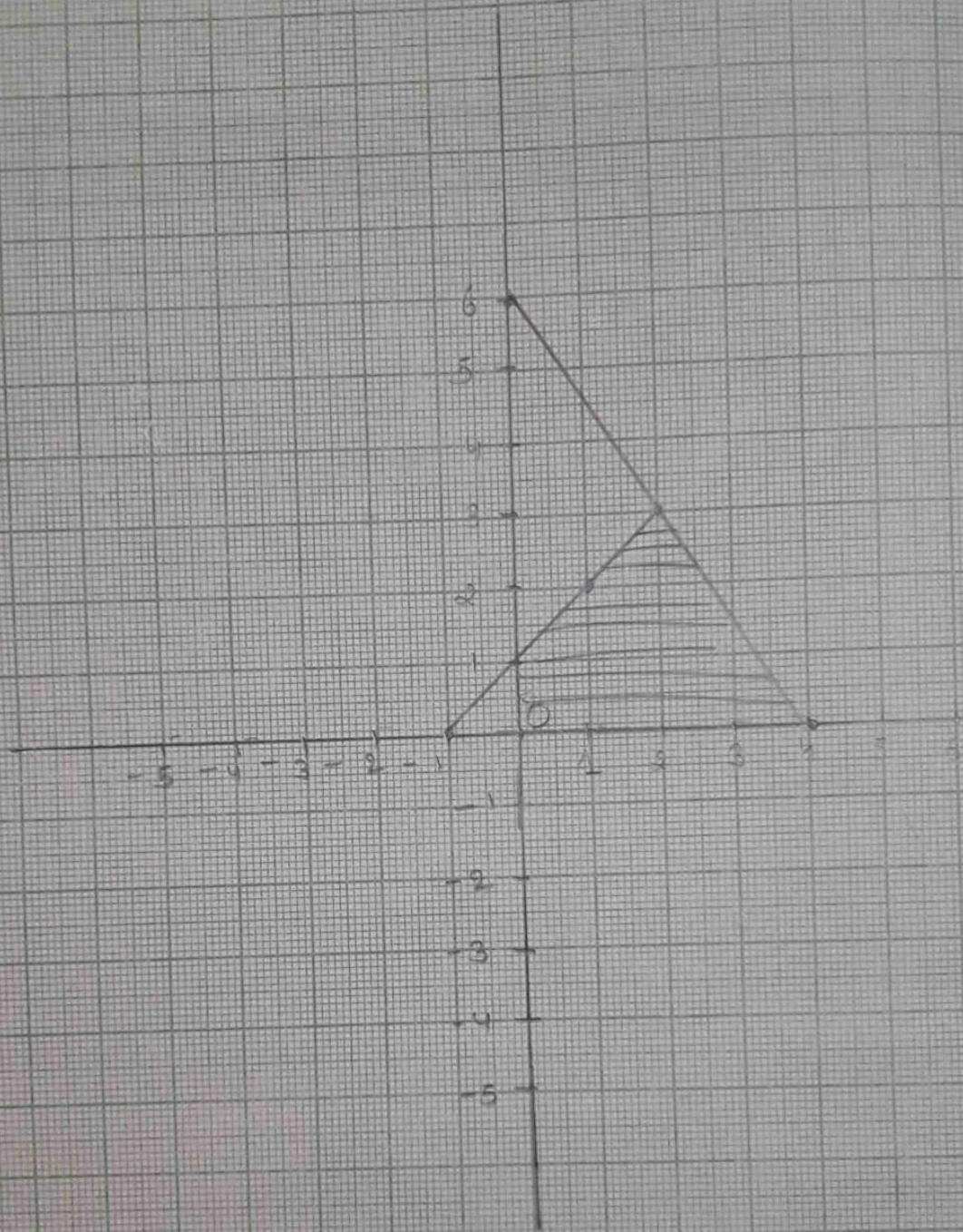
u	0	-2	4
y	6	3	0

$$u = 2$$

$$y = \frac{-6 + 12}{2} = 3$$

$$u = 4$$

$$y = \frac{-12 + 12}{2} = 0$$



= Area of triangle

$$\frac{1}{2} \times B \times H$$

$$\frac{1}{2} \times 5 \times 3$$

$$= \frac{1}{2} \times 15 = 7.5 \text{ sq. unit}$$

Q34

$$p(u) = 2u^3 - 4u - u^2 + 2$$

two zeros are $\sqrt{2}, -\sqrt{2}$

$$\Rightarrow (u - \sqrt{2})(u + \sqrt{2}) = u^2 - 2$$

$u^2 - 2$ is the factor of $p(u)$

Apply the division Algorithm

$$P(u) \quad 2u^3 - 4u - u^2 + 2 \quad \div \quad u^2 - 2$$

remainder $r(u) = 0$

$$p(u) = q(u) \times g(u)$$

$$\therefore 2u^3 - 4u - u^2 + 2 = (u^2 - 2)(2u - 1)$$

$$= (u - \sqrt{2})(u + \sqrt{2})(2u - 1)$$

$$(u - \sqrt{2})(u + \sqrt{2})(2u - 1) = 0$$

$$\Rightarrow u = \sqrt{2} \quad u = -\sqrt{2} \quad , \quad u = \frac{1}{2}$$

Q35

$$(4, -3) \quad 3:1 \quad (6, 5)$$

$$P(x, y) = \left(\frac{m_1 x_2}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right)$$

$$x = \frac{3(6) + 1(4)}{3+1} = \frac{24+4}{4} = \frac{28}{4} = 7$$

$$y = \frac{3(5) + 1(-3)}{3+1} = \frac{15-3}{4} = \frac{12}{4} = 3$$

$P(7, 3)$ is the required point

Q36

(I) $(0, 10)$ (II) $(3, 5)$ (III) $4m$

Q37

(I) Parabola

(II) $0, 1$ (III) $-3, 2$

Q38.

(i) 24 cm^2

(ii) - RHS

(iii) - $16:81$